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Guided filter and convolutional network based tracking for infrared dim moving target

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HIGHLIGHTS

- An infrared dim-small target tracking based on guided image filter and two-layer convolutional network is presented.
- A discriminative template feature is obtained via the guided filter.
- A simplified convolution neural network is adopted to obtain a powerful target representation.
- The Bayesian classifier with an adaptive learning rate does well in the classification of image templates.
- The particle filter provides a rigorous general framework for dynamic state estimation problems.

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ABSTRACT

The dim moving target usually submerges in strong noise, and its motion observability is debased by numerous false alarms for low signal-to-noise ratio. A tracking algorithm that integrates the Guided Image Filter (GIF) and the Convolutional neural network (CNN) into the particle filter framework is presented to cope with the uncertainty of dim targets. First, the initial target template is treated as a guidance to filter incoming templates depending on similarities between the guidance and candidate templates. The GIF algorithm utilizes the structure in the guidance and performs as an edge-preserving smoothing operator. Therefore, the guidance helps to preserve the detail of valuable templates and makes inaccurate ones blurry, alleviating the tracking deviation effectively. Besides, the two-layer CNN method is adopted to obtain a powerful appearance representation. Subsequently, a Bayesian classifier is trained with these discriminative yet strong features. Moreover, an adaptive learning factor is introduced to prevent the update of classifier's parameters when a target undergoes severe background. At last, classifier responses of particles are utilized to generate particle importance weights and a re-sample procedure preserves samples according to the weight. In the predication stage, a 2-order transition model considers the target velocity to estimate current position. Experimental results demonstrate that the presented algorithm outperforms several relative algorithms in the accuracy.

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1. Introduction

The infrared target tracking has been widely used in military and civilian. The main task is to estimate the target location in the next frame. When the distance between the optical system and a moving target is large, the target in the imaging plane only contains only several pixels and has low Signal-to-Noise Ratio (SNR). With strong radiation, dim moving targets usually submerge in cluttered backgrounds, which is the crucial challenge. Besides, the motion state of a small target can be random and uncertain

when it moves fast. Therefore, the accurate detection and tracking of a dim target is considered as a difficult work [1–15].

In general, infrared dim target tracking algorithms can be divided into two categories. One is based on the image filtering [4–11] and another is based on the target modeling and matching [12–15]. The first approach establishes an ideal motion model and then detects the target by filtering algorithms, such as temporal-spatial fusion filter [4], Moving Pipeline Filter (MPF) [5], Correlation Filter (CF) [6,7] and Particle Filter (PF) [8–11]. In CF-based algorithm [6], the frequency response of a filter bank was varied in each frame accordingly with local statistics of the observed signal. Afterwards, the target was detected in the restored frame by analyzing the correlation intensity planes obtained at the output of the filter bank. However, tracking deviation is frequent as the

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estimation of the dynamic background is inaccurate. In [10], Wang et al. utilized Markov chain and Monte Carlo methods to approximate the solution of sequential target's state estimation. The particle filtering is useful for nonlinear motion. Nevertheless, it is computationally intensive, and also faces convergence problems over complex backgrounds. In addition, Li et al. presented a particle filter-based tracking algorithm to copy with the uncertainty of the dim moving target tracking [11]. The discriminative over-complete dictionary was utilized to enlarge the difference between the target particle and the background particle. Further, the particle discriminative sparse representation can improve the accuracy of target motion estimation by means of heightening the weight of target particles. Another type is achieved by modeling and matching the target feature, such as Template Matching Tracking (TMT) [12,13], optical flow [14,15] and Mean-Shift (MS) [16–18]. The TMT algorithm completes matching between frames using the gray feature. In [13], an improved TMT algorithm was suggested based on the statistics and the normalize cross-correlation algorithm. Equally popular are the MS algorithm [17,18], which utilizes a basic shape to represent the target and uses the histogram as a main target descriptor. The MS algorithm is computationally efficient and effective for tracking the position and orientation of a target. Yet, gray statistics may be unstable as illumination changes, and complex backgrounds make the target feature less discriminative.

Moreover, a tracking mechanism based on an online training classifier was presented. Zhang et al. presented a Fast Compressive Tracking (FCT) algorithm [19], which was formulated as a binary classification via a Bayesian classifier. However, it is difficult to pick out the target template when the small target submerges in the background clutter. In order to track an infrared target, Li et al. combined the appearance model in the FCT algorithm with the effective ℓ_1 algorithm [20]. Nevertheless, the feature information is insufficient for an infrared dim target. Recently, Wang et al. [21] used a two-layer Convolutional Neural Network (CNN) to learn hierarchical features from auxiliary video sequences. Yet, the offline training is time-consuming. In [22], Zhang et al. presented a simplified CNN to obtain robust representations for visual tracking. The method was proved to have a good performance in feature extraction. Therefore, it is introduced into the infrared dim target tracking field for better target representations.

In this paper, an infrared target tracking framework, which integrates the Guided Image Filter (GIF) [23] and the simplified CNN [22] into the particle filter [24] is presented to deal with the uncertainty of the dim moving target over several kinds of cluttered backgrounds. A typical tracking system consists of three components [25], i.e., an appearance model which is the most important factor, a motion model which needs to be updated, and a search strategy for finding the most likely location in the current frame. To represent the target, the holistic image intensity is extracted from the object region. To restrain the negative influence of the background clutter, target templates and candidate templates are processed by the guided filter. As an appearance model, the GIF algorithm considers the appearance information between the guidance and candidate images. Therefore, candidate templates can be filtered to be more smooth because of the high similarity, or to be blurry due to the background edge or the broken background. Moreover, a simplified two-layer convolutional network is adopted to obtain an effective target representation. As a search strategy, the maximal response of the particle filter is utilized to determine the target template. Besides, an adaptive learning factor in the Bayesian classification is adopted to control whether classifier parameters are updated when targets submerge in cluttered backgrounds. As a motion model, the particle filter framework is adopted to predict possible targets. In the predication stage, a 2-order transition model considers the target velocity to estimate

the current position. Experimental results validate that the presented algorithm can significantly improve the accuracy of target tracking over infrared image sequences, in comparison with several baseline algorithms.

The remainder is organized as follows. In Section 2, the gray distribution of the infrared image is analyzed firstly and the procedure of the Bayesian classification is reviewed. In Section 3, an improved method based on the GIF and the CNN is introduced in detail. In Section 4, the performance of the proposed algorithm is verified over several infrared image sequences. Finally, conclusions are drawn in Section 5.

2. Related work

2.1. Infrared image gray distribution

In an infrared sequence with the cluttered background (see Fig. 1), a dim moving target has the probability to appear at any position. Based on the gray distribution of an infrared image, candidate templates can be divided into five gray structures, including *target-in-cloud* template, *clean-sky* template, *middle-cloud* template, *edge-cloud* template and *broken-cloud* template, which are shown in Fig. 2. Besides, three dimensional views of these templates are shown in Fig. 3.

As shown in Fig. 2(a), the *target-in-cloud* template is recognized as a positive sample and the target region has a larger intensity than its neighbor pixels (see Fig. 3(a)). The other four types are all negative samples, i.e., template 2–5. The *clean-sky* template, which has low gray intensity (see Fig. 3(b)), is obtained in the background with litter clutter. The *middle-cloud* template denotes the cloud background, and the intensity value tends to be larger than that in the *clean-sky* template. As shown in Fig. 3(d), there is a clear boundary in the *edge-cloud* template, which has similar frequency characteristic to a dim target. Finally, the *broken-cloud* template occurs when the infrared array has blind pixels.

2.2. Bayesian classification

The effective Bayesian classification is utilized to choose the target template with the maximum probability. Given a sample set whose elements contain two types of samples, i.e., positive samples and negative samples. For each sample, the representation of its gray feature is $\mathbf{v} = (v_1, \dots, v_m)$, $v_i (i = 1, 2, \dots, m)$ represents the image intensity of the i -th pixel in each sample image and m denotes the pixel number. All elements in $\mathbf{v}$ are independently distributed and the naive Bayesian classifier is represented by

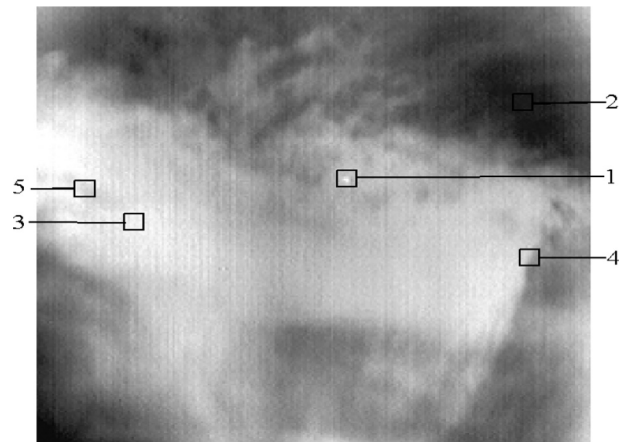


Fig. 1. Five types of templates.



Fig. 2. Candidate templates. (a) 1 target-in-cloud (b) 2 clean-sky (c) 3 middle-cloud (d) 4 edge-cloud (e) 5 broken-cloud.

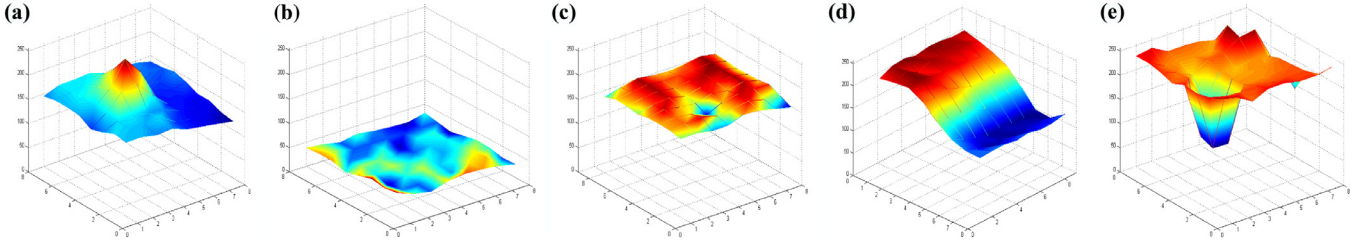


Fig. 3. 3D views of candidates in Fig. 2. (a) 1 target-in-cloud (b) 2 clean-sky (c) 3 middle-cloud (d) 4 edge-cloud (e) 5 broken-cloud.

$$H(v) = \log \left(\frac{\prod_{i=1}^m P(v_i|y=1)p(y=1)}{\prod_{i=1}^m P(v_i|y=0)p(y=0)} \right) \quad (1)$$

$$= \sum_{i=1}^m \log (P(v_i|y=1)/P(v_i|y=0))$$

where $\prod$ represents the product operator and $\log()$ denotes the logarithmic function. The equation $P(y=1)=P(y=0)$ is the uniform prior, and $y = \{0,1\}$ is a binary variable which represents the sample label, 'y = 1' represents the target sample and 'y = 0' represents the negative sample class. Moreover, the conditional distribution $P(v_i|y)$ in the classifier $H(v)$ are modeled as Gaussian distributed with four classifier parameters $\mu_i^1, \sigma_i^1, \mu_i^0, \sigma_i^0$, which will be updated with learning parameter λ .

$$P(v_i|y=1) \sim N(\mu_i^1, \sigma_i^1) \quad (2)$$

$$P(v_i|y=0) \sim N(\mu_i^0, \sigma_i^0) \quad (3)$$

The scalar parameters in Eqs. (2) and (3) are incrementally updated by

$$\mu_i^1 \leftarrow \lambda \mu_i^1 + (1 - \lambda) \mu^1 \quad (4)$$

$$\sigma_i^1 \leftarrow \sqrt{\lambda (\sigma_i^1)^2 + (1 - \lambda) (\sigma^1)^2 + \lambda (1 - \lambda) (\mu_i^1 - \mu^1)^2} \quad (5)$$

$$\mu_i^0 \leftarrow \lambda \mu_i^0 + (1 - \lambda) \mu^0 \quad (6)$$

$$\sigma_i^0 \leftarrow \sqrt{\lambda (\sigma_i^0)^2 + (1 - \lambda) (\sigma^0)^2 + \lambda (1 - \lambda) (\mu_i^0 - \mu^0)^2} \quad (7)$$

where $\lambda > 0$ is a learning factor, μ_i^1 and σ_i^1 are the mean value and the variance of a target template's feature $\mathbf{v}$. μ_i^0 and σ_i^0 are the mean value and the variance for a negative sample.

3. Proposed tracking algorithm

3.1. Guided image filter

Generally, infrared dim target tracking is interfered by the *edge-cloud* template primarily, as well as the *broken-cloud* template, due to the fact that both of them have a similar frequency characteristic to the dim-small target. In order to solve the problem, a suitable

image filter is adopted to enlarge the difference between target templates and types of negative templates.

With a local linear model, the Guided Image Filter (GIF) algorithm [19] generates an output by considering the content of a guidance image. The filtering process is general linear variable, which includes an input image p , a guidance image G and an output image q . Assuming that q is a linear transform of G in a window w_k , which centered at the pixel k .

$$q_i = a_k G_i + b_k, \quad \forall i \in w_k \quad (8)$$

where a_k and b_k are linear coefficients assumed to be constant in a box w_k with the radius being r . This local linear model ensures that q has an edge only if G has an edge, because $\nabla q = a \nabla G$, where ∇ denotes the image gradient.

To determine linear coefficients, a cost function E that minimizes the difference between p and the filtered result q is represented by

$$E(a_k, b_k) = \sum_{i \in w_k} ((a_k G_i + b_k - p_i)^2 + \varepsilon a_k^2) \quad (9)$$

where ε is a regularization parameter, and $\sum$ denotes the sum operator. The solution can be obtained by minimizing Eq. (9) with linear regression as follows.

$$a_k = \left(\sum_{i \in w_k} G_i p_i - \mu_k \sum_{i \in w_k} p_i \right) / (|w| (\sigma_k^2 + \varepsilon)) \quad (10)$$

$$b_k = \sum_{i \in w_k} p_i / |w| - a_k \mu_k \quad (11)$$

where μ_k and σ_k^2 are the mean and the variance of the guidance G in w_k , $|w|$ is the pixel number in w_k .

The detail that how the GIF keeps the edge-preserving smoothing property can be found in [23]. Similar to the method in [26], the GIF algorithm is introduced into the infrared dim target tracking filed for better distinguishing the target templates from distractive ones. To avoid being influenced by the background clutter, the target template in the initial frame is regarded as the guidance image, which is shown in Fig. 4. Moreover, the local window radius r is set as 3 and ε is 0.04. The effect of the GIF algorithm is shown in Figs. 5 and 6.

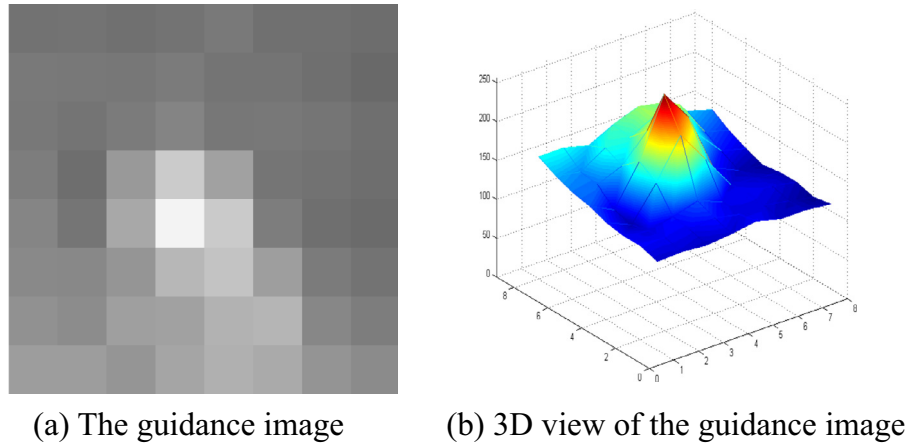


Fig. 4. The guidance image.

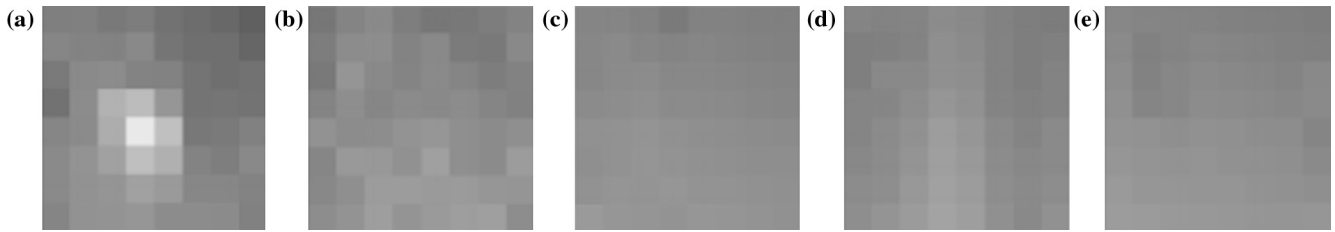


Fig. 5. Corresponding outputs via GIF. (a) 1 target-in-cloud (b) 2 clean-sky (c) 3 middle-cloud (d) 4 edge-cloud (e) 5 broken-cloud.

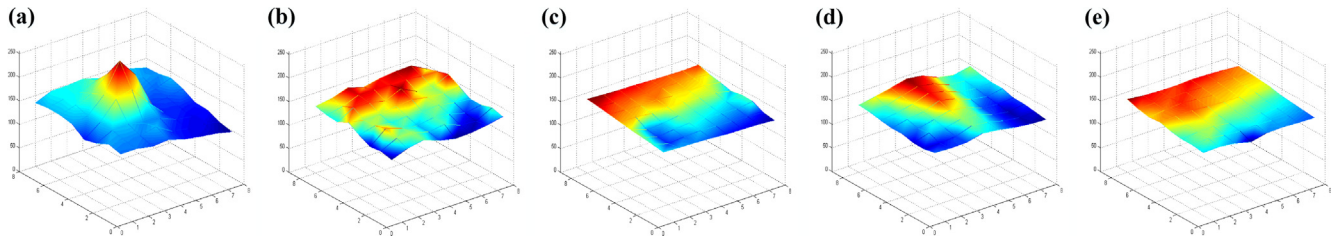


Fig. 6. 3D views of Fig. 5, orderly. (a) 1 target-in-cloud (b) 2 clean-sky (c) 3 middle-cloud (d) 4 edge-cloud (e) 5 broken-cloud.

As shown in Fig. 5(a), the filtered target template resembles the original target template (see Fig. 2(a)). In Fig. 6(b)–(d), the filtered result of the other four templates come to be smooth or blurry by considering the content of the guidance image, which is because that the output has an edge only if the guidance image has an edge. Besides, the boundary in the *edge-cloud* template is not clear at all in Fig. 6(d). Therefore, these outputs can be either classified by discriminative algorithms or matched through the classic algorithm in an effective way. However, the simple gray value may be insufficient for dim targets to be distinguished from the cluttered background that is strong.

3.2. Convolutional network based object representation

Motivated by the success of convolution network for visual tracking [22], an improved generative model that considers the local structural information of a dim moving target is utilized for object representation. Given a target template in the positive template set A_+ that has been processed by the guided filter, a hierarchical representation architecture with a convolutional network contains two separated layers. First, local selective features are extracted from a bank of filters convolving the input image at each

position. Second, selective features are stacked together to form a representation. The detail is shown in Figs. 7 and 8.

As shown in Fig. 7, each candidate template is processed by the GIF algorithm, and warped to a canonical size of $n \times n$ pixels (e.g., 12×12 pixels). The filtered target template is represented by image intensity, denoted as I_j ($j = 1, 2, \dots, N_p$) $\in R^{n \times n}$, where N_p is the number of the target template. Moreover, a set of overlapping local image patches Y_i ($i = 1, 2, \dots, l$) centered at each pixel position is densely sampled inside the filtered template through sliding a window of size $w \times w$, where $Y_i \in R^{w \times w}$ is the i -th image patch and $l = (n - w + 1) \times (n - w + 1)$.

3.2.1. Simple layer

Subsequently, N_p patches are sampled as fixed filters to extract the template features at the first frame. Given the filter $Y_i \in R^{w \times w}$, its response on the filtered image I_j is denoted with a feature map $S_i \in R^l$, which can be expressed as

$$S_i = Y_i \otimes I_j \quad (12)$$

where $\otimes$ is the convolution operator.

As shown in Fig. 7, some simple cell feature maps are obtained. Simple cell feature maps of three target templates have similar

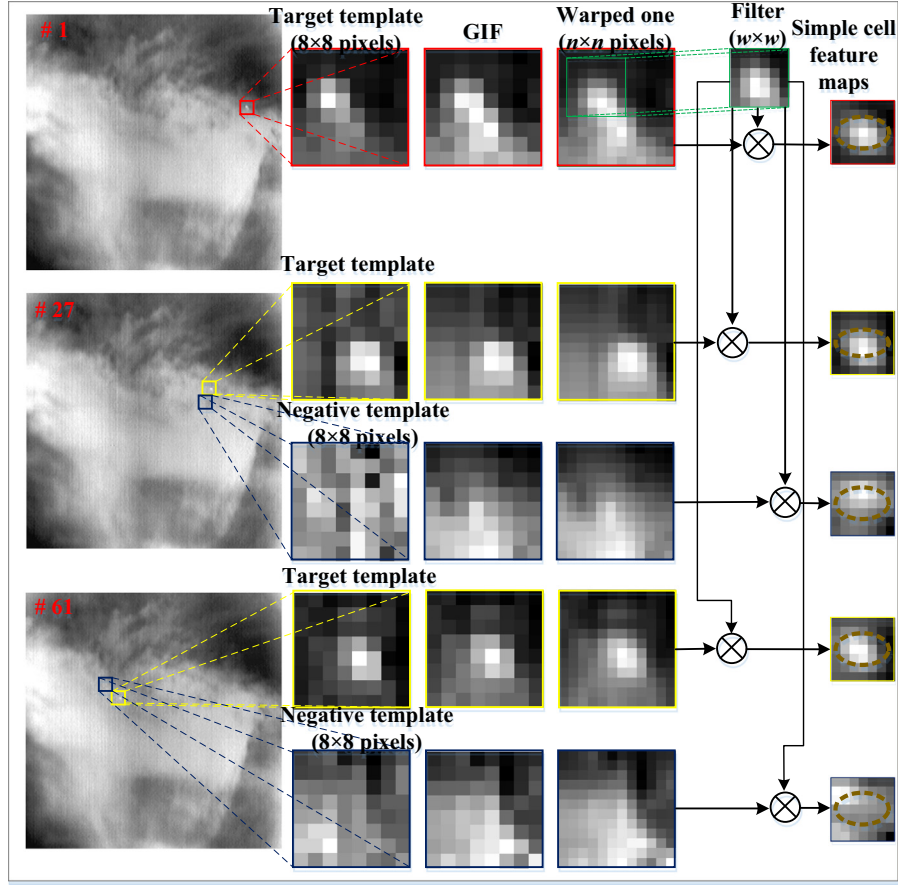


Fig. 7. Simple cell feature maps.

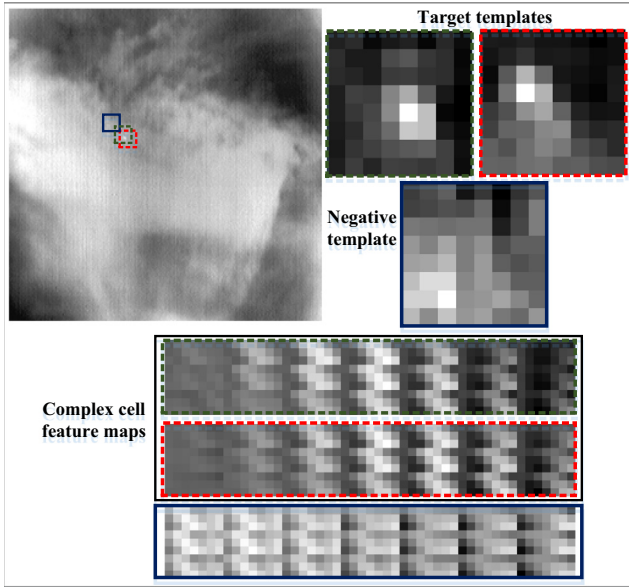


Fig. 8. Complex cell feature maps.

geometric layout despite appearance variation, demonstrating that the local filter Y_i can extract useful information across entire image templates. Therefore, the global geometric layout information of the dim target template can also be effectively exploited. For negative templates that contain cluttered background, both feature maps undergo larger difference. In short, the selective filter is localized that can extract local property of a dim moving target.

3.2.2. Complex layer

Each simple cell feature map S_i simultaneously encodes the local structural information of the target template, generating a good representation to handle appearance variations. To enhance the strength of this representation, a three-dimensional tensor $C \in R^{(n-w+1) \times (n-w+1) \times N_p}$, which stacks N_p different simple cell feature maps, is constructed to obtain a complex cell feature map. The illustration of obtaining complex cell feature maps is shown in Fig. 8.

As shown in Fig. 8, the two target templates have a similar complex cell feature map (inside the black rectangle), and the negative template undergoes large difference in the complex cell feature map. Therefore, the property makes the Bayesian classification more stable in calculating the response.

3.3. Adaptive learning factor for Bayesian classification

The candidate template with a maximal classifier response is marked as a target template. Supposing that all responses values H are less than 0, the target template is actually a negative sample. When the target moves across the thick cloud, most of the target templates contain strong edge or cluttered background. To avoid being influenced by these bad candidate templates, an suitable way is to stop the update of scalar parameters when the target submerges in the cloud.

Afterwards, an adaptive learning factor λ_1 of the Bayesian classifier is presented to cope with the model update. Besides, the maximal response $H_{\max}$ is defined as

$$H_{\max} = \max(H(v)) \quad (13)$$

where $\max()$ denotes a function of calculating the maximum.

Therefore, the classifier needs no update on account of that the maximal response is less than 0, and the learning factor λ_1 of the classifier is set as 1. Otherwise, it is set as 0.95.

3.4. Particle filter for tracking

Once the tracker location at frame t is updated, the classification only considers the candidate template with the maximal classifier response as the target at frame $t + 1$. Supposing indistinguishable templates that are similar to the dim moving target template is contained in candidate samples, an inaccurate classifier response may be obtained, leading to tracking deviation. Moreover, within a short period, the motion state of the dim target can be random and uncertain, which acts like non-linear and non-Gaussian. Therefore, the particle filter [9,24], which considers the tracking task as dynamic estimation of particles' state, is utilized to improve the performance of the classification.

3.4.1. The particle filter framework

The essentially step is prediction and update. Supposing the required Probability Density Function (PDF) $p(x_{t-1}|z_{1:t-1})$ at time $t-1$ is available, the prior probability density $p(x_t|z_{1:t-1})$ at time t is predicted by

$$p(x_t|z_{1:t-1}) = \int p(x_t|x_{t-1})p(x_{t-1}|z_{1:t-1})dx_{t-1} \quad (14)$$

where $z_{1:t-1}$ denotes the cumulative set of measurements up to previous instant, and $p(x_t|x_{t-1})$ is the state transition probability density to predict the prior at time t .

Given the observation z_t at time t , the goal is to determine the posterior PDF $p(x_t|z_{1:t})$ using Bayes theorem

$$p(x_t|z_{1:t}) = p(z_t|x_t)p(x_t|z_{1:t-1})/p(z_t|z_{1:t-1}) \quad (15)$$

where $p(z_t|x_t)$ is described by the observation equation and $p(z_t|z_{1:t-1})$ denotes the normalizing denominator.

Therefore, the PF algorithm provides an approach to approximate the optimal Bayesian solution of the recurrence relations in Eqs. (14) and (15), which is intractable in most cases.

At time $t-1$, a set of random samples $\{x_{t-1}(k)\}_{k=1}^N$ with importance weights w_{t-1}^k are obtained from the PDF $p(x_{t-1}|z_{1:t-1})$ of the dim target's state. Then, the PF algorithm propagates and updates these samples to obtain a set of new samples $\{x_t(k)\}_{k=1}^N$, which are approximately distributed as $p(x_t|z_{1:t})$.

In the prediction stage, a new samples $x_t^*(k)$ at time t is obtained by Eq. (14) with each sample $x_{t-1}(k)$. On receipt of the measurement z_t , the likelihood of each prior sample is evaluated and a normalized weight q_k for each predicated sample is expressed by

$$q_k = p(z_t|x_t^*(k)) / \sum_{j=1}^N p(z_t|x_t^*(j)) \quad (16)$$

Then, the prior samples are re-sampled to generate an unweighted particle set according to the importance weights, helping to avoid degeneracy.

3.4.2. Transition model and observation model

Since a dim target has no texture information, the sample state vector can be defined as $x = (x_p, y_p)$, where x_p and y_p represent the location of the sample. To initial the tracker, N samples' $x_0(k)$ ($k = 1, 2, \dots, N$) are drawn from the initial PDF $p(x_0|z_0) \equiv p(x_0)$, where z_0 means no measurements, $p(x_0)$ is the prior of the state and is obtained by user input.

In the prediction stage, the samples are propagated by the transition model. The 2-order, auto-regressive model which considers the previous two states of each sample is utilized to fuse the velocity information of the dim moving target. The model is express as

$$x_t(k) = 2x_{t-1}(k) - x_{t-2}(k) + \eta \quad (17)$$

where $x_t(k)$ denotes the state vector of the k -th particle at time t , and η represents the covariance of a two-dimension, white-noise with zero-mean, which is independent of past and current states with two different standard deviations for each dimension.

In the update stage, the observation models in Eq. (16) to estimate the weights for each sample is represent by

$$p(z_t|x_t^*(k)) \propto \exp(H(v^{(k)})) \\ = \prod_{i=1}^{I \times N_p} \frac{p(v_i^{(k)}|y=1)}{p(v_i^{(k)}|y=0)} \quad (18)$$

where $v^{(k)}$ denotes the feature vector of the k -th particle, which has been processed with the GIF algorithm and the CNN algorithm.

As weights of most particles are negligible, the prior particle set will be re-sampled to avoid the degeneracy. The remained particle set retains particles whose classifier response are relatively high, and contains a redistribution of particles. In the presented GIF-CNN algorithm, these re-sampled particles will work together to find the most possible sample in the next frame, which reduces tracking deviation even if samples are poor.

3.4.3. The GIF-CNN algorithm

In summary, the GIF-CNN algorithm for dim moving target tracking takes advantages of a discriminative yet strong feature and a particle filtering framework. The Schematic is shown in Fig. 9. The main steps of the GIF-CNN algorithm are summarized in Algorithm 1.

Algorithm 1. GIF-CNN Tracker

Input t -th Infrared image frame, target template set A_+ at the initial frame

Step 1: Sampling two sets of image patches, $Z^\alpha = \{z | \|I(z) - I_t\| < \alpha\}$ and $Z^{\beta,0} = \{z | \beta < \|I(z) - I_t\| < \theta\}$ with $\alpha < \beta < \theta$.

Step 2: All candidates are processed by the GIF algorithm, and then normalized to the same size, i.e., 12×12 pixels. Besides, a filter set is extracted from the normalized target template set at the initial frame.

Step 3: At the frame $t + 1$, sampling abundant particles which are processed by the GIF algorithm. Obtaining their complex cell feature maps and concatenating all the elements to obtain a feature vector.

Step 4: Predicating particles by Eq. (14) with the transition model in Eq. (17).

Step 5: Updating particles by Eq. (15) with the observation model in Eq. (18), and finding the particle with some largest weights. Target location is marked by the center of these particles.

Step 6: Re-sampling particles to retain useful templates with a high weight.

Step 7: Sampling two sets of image patches as **Step 1**. Moreover, discriminative feature vectors are obtained similar to **Step 3**.

Step 8: Training the Bayesian classifier and updating the classifier parameters with an adaptive learning factor according to Eq. (13)

Output: Tracking location and classifier parameters.

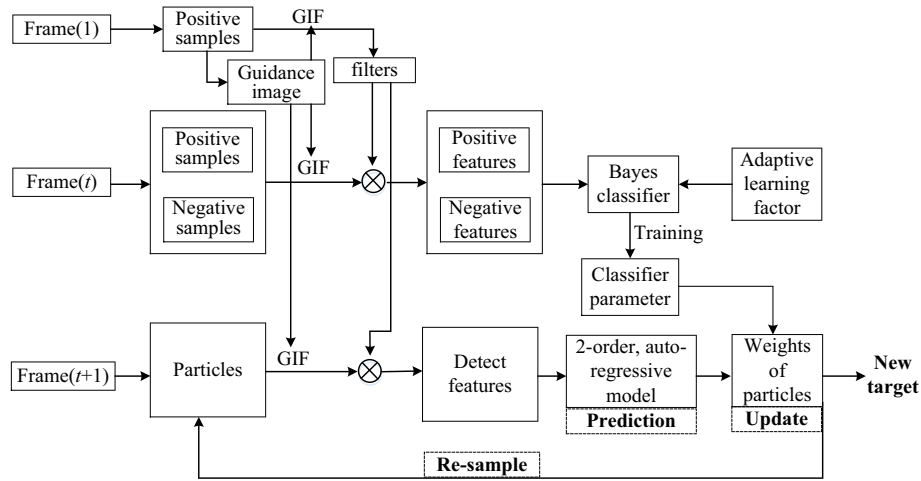


Fig. 9. Schematic of the presented algorithm.

4. Experimental results and analysis

The performance of the GIF-CNN algorithm is evaluated over five groups of infrared sequences containing different backgrounds, which are collected from diverse sources, e.g., image sequences captured by us, dataset from the Object Tracking and Classification in and Beyond the Visible Spectrum (OTCBVS) benchmark [27], etc. Seq.1–5 are named *plane1*, *plane2*, *plane3*, *plane4* and *tank*, respectively, as shown in Fig. 10. The detail of these infrared image sequences and small targets is described in Table 1.

As shown in Fig. 10, Seq.1–2 are obtained by capturing the central 256×256 pixels from images taken by a 256×256 pixels uncooled IRFPA camera. As shown in Table 1, Seq.1 consists of 86 frames, a small target submerges in the bright background. Specially, it moves across the edge of cloud in some images. Seq.2 is in background of strong edge, and the dim target's SNR is no more than 1.5. Besides, the target size is $2 \times 2 \sim 4 \times 4$ pixels and the sequence length is 120.

Seq.3 is obtained by capturing the central 400×450 pixels from images taken by infrared camera. This sequence contains a target with the speed of about 8 pixels per frame, and its length is 113.

Seq.4 is obtained from the OTCBVS benchmark, a dim target under the space background appears in the upper right corner at

the beginning and follows a downward diagonal path. Besides, some false objects appears in the image sequence. The sequence includes 200 frames.

In Seq.5 (size 200×200 pixels), the target motion is unstable, which is caused by camera shake. Its size is from 3×3 pixels to 7×7 pixels.

Experiments are implemented using MATLAB 2014a on a PC with a 3.3 GHz Intel-Core CPU and 4 GB memory. The GIF-CNN algorithms in Algorithm 1 are tested and compared with five algorithms, including the MPF algorithm [5], the improved TMT algorithm [13], the MS-PF algorithm [8], the FCT algorithm [19] and the ℓ_1 algorithm [20].

4.1. Experimental setup

The parameter can be set as follows. The size of the tracking box is 8×8 pixels (its center denotes the target location), for the reason that the small target can be contained completely, and it should not be too large in order to prevent the background clutter. In the GIF algorithm, the radius r of a square window w_k is 3 and the balancing parameter ε is fixed as 0.04. In the CNN method, the size of the warped image is set to 12×12 ($n = 12$), the filter size w is set as 6 and the number of the filter is set to be 9. In the

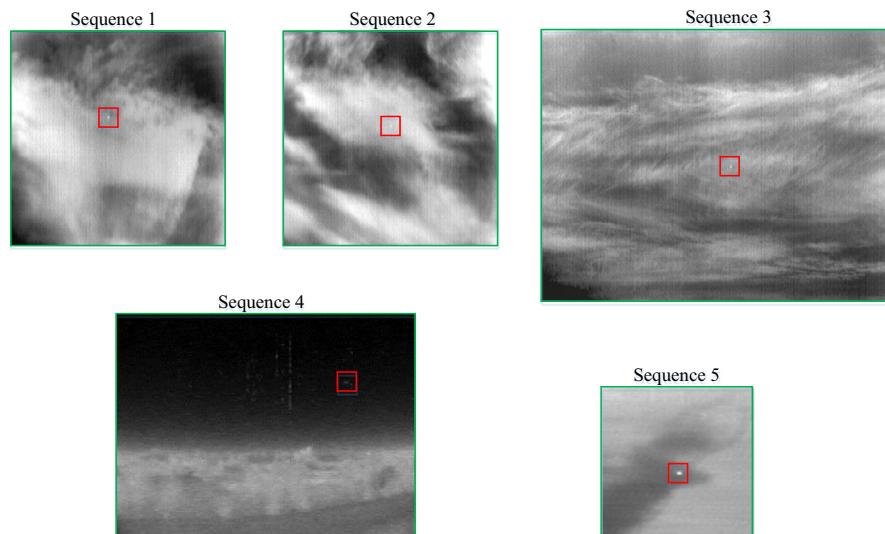


Fig. 10. The five real infrared image sequences.

Table 1
Overview of test infrared sequences.

Sequence	Frames	Image size	Target size	Description
Seq.1	86	256 × 256	2 × 2	An airplane with bright sky background
Seq.2	120	256 × 256	3 × 3	An airplane with strong cloud edge
Seq.3	113	400 × 450	3 × 3	An airplane with large image size
Seq.4	200	240 × 320	3 × 3	Plane with low gray value, sever background cloud
Seq.5	200	128 × 128	5 × 5	A tank under the ground background

Bayesian classification, α and β are set as 3 and 15, respectively. Both the initial learning parameter λ and λ_1 are 0.95. In the particle filter, the two-dimension standard deviation η is set as $[0.5, 0.5] \sim [5, 5]$, referring to the target velocity. The particle number is defined as 100, as a trade-off between computation efficiency and effectiveness.

4.2. Qualitative comparison

Qualitative results of six groups of tracking algorithms are presented in Figs. 11–15, and particles are shown in Fig. 16.

In the *plane1* sequence, the background undergoes strong radiation. Both the ℓ_1 algorithm and the MPF algorithm undergo large deviation at frame #85, which is shown in Fig. 11(d). The TMT algorithm drifts to the background when the dim target moves into the bright cloud, e.g., frame #36, #53 and #86. The MS-PF algorithm fails to detect the target at frame #36 and the tracking deviation cannot be eliminated. Only the FCT tracker and the GIF-CNN tracker track the target well in all images.

In the *plane2* sequence, the edge of cloud is strong. Only the GIF-CNN algorithm gives an accurate estimation of the target's position at frame #119. As shown in Fig. 12, the MPF algorithm is sensitive to the edge of cloud and always loses the target, for the reason that the slow-moving edge easily forms a pipeline and is detected as a target. The TMT algorithm fails to track the target when it moves cross the strong edge and the problem is unrecoverable (e.g., frame

#119). The FCT algorithm has tracking deviation prematurely (e.g., frame #86), which is because that classifier responses of candidate templates are influenced by strong edge. In the MS-PF algorithm, the gray statistics of the image patch is utilized to obtain particle weights, and it is obvious that tracking deviation occurs once the dim target submerges in the cloud (e.g., frame #86 and #119).

In the *plane3* sequence, the image size is large and the target moves fast. The MPF fails to detect the target at frame #62 (see Fig. 13(c)). In the *plane4* sequence, the background has a low gray value and the space target disappears behind the horizon. The tracking result of the MPF is unacceptable. Both the TMT algorithm and the FCT algorithm lose the dim target after frame #120, and a false target is mistaken for a true target at frame #200.

In the *tank* sequence, the camera has a sever shake. The MPF algorithm has frequent detection error (e.g., frame #148 and #190), caused by continuous motion of background pixels. The TMT tracker may drift to background, yet, it is possible to detected the target again once the target goes into the target box. Benefiting from the particle filter, the MS-PF algorithm and the GIF-CNN algorithm can obtain a target template with a maximal weight from the particle set. The FCT algorithm can track the target well, due to the fact that the feature of a target with a larger size can be discriminative and the ground background contains less clutter.

As shown in Fig. 16, each sampled particle (the yellow rectangle) around the target has a certain weight, helping to obtain an optimal estimation (the red rectangle) in the current frame.

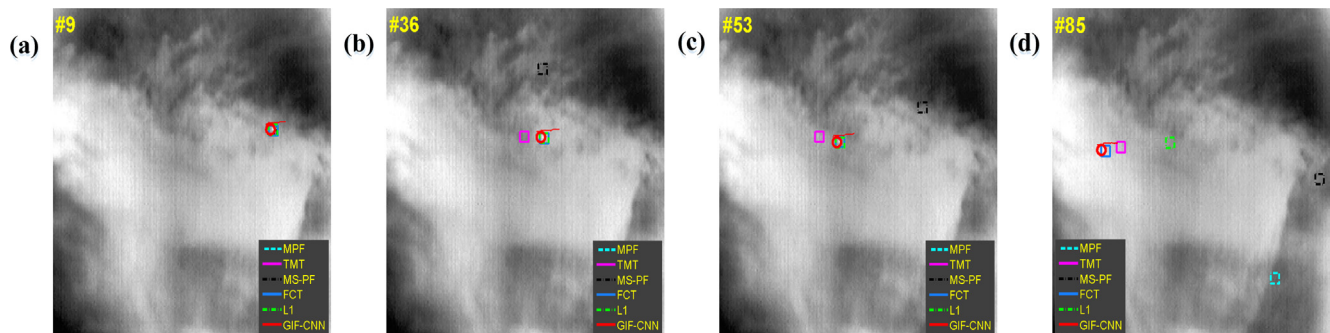


Fig. 11. Qualitative results of the 6 trackers over *plane1*, with bright background. Best viewed on color display.

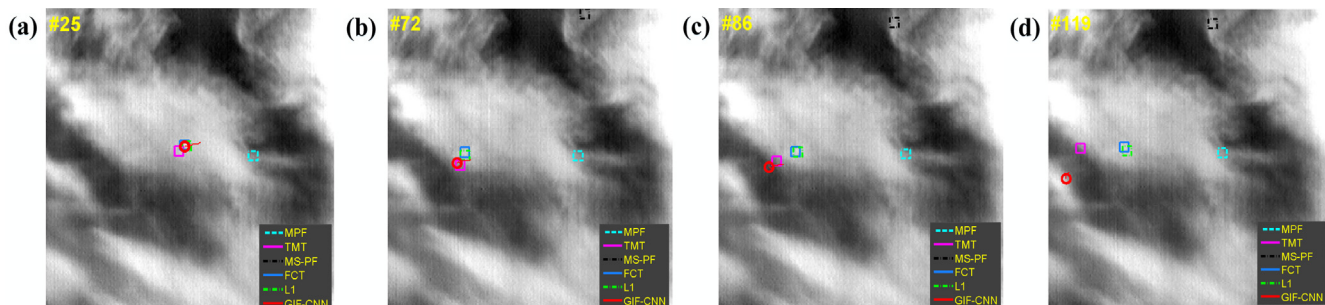


Fig. 12. Qualitative results of the 6 trackers over *plane2*, in which the edge is strong. Best viewed on color display.

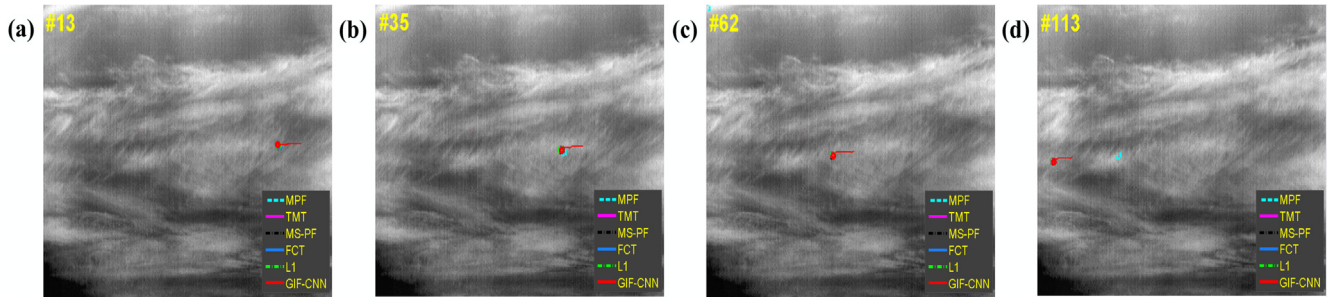


Fig. 13. Qualitative results of the 6 trackers over *plane3*, with large image size. Best viewed on color display.

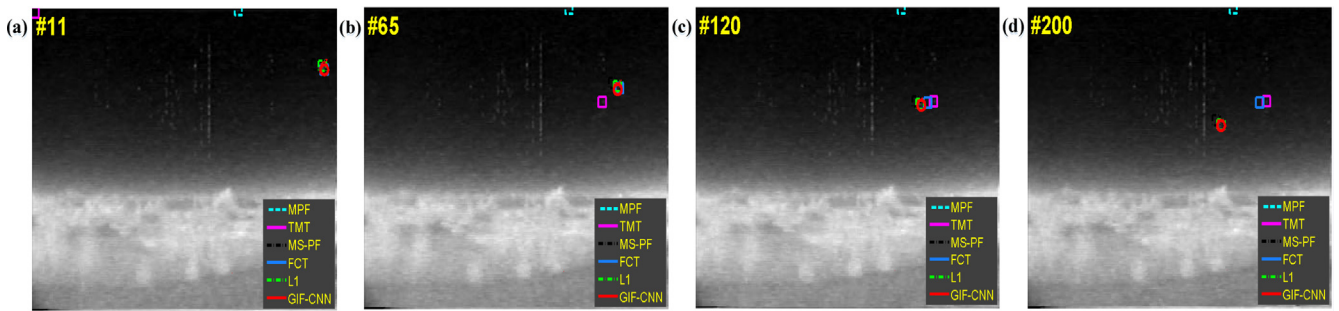


Fig. 14. Qualitative results of the 6 trackers over *plane4* with low gray value. Best viewed on color display.

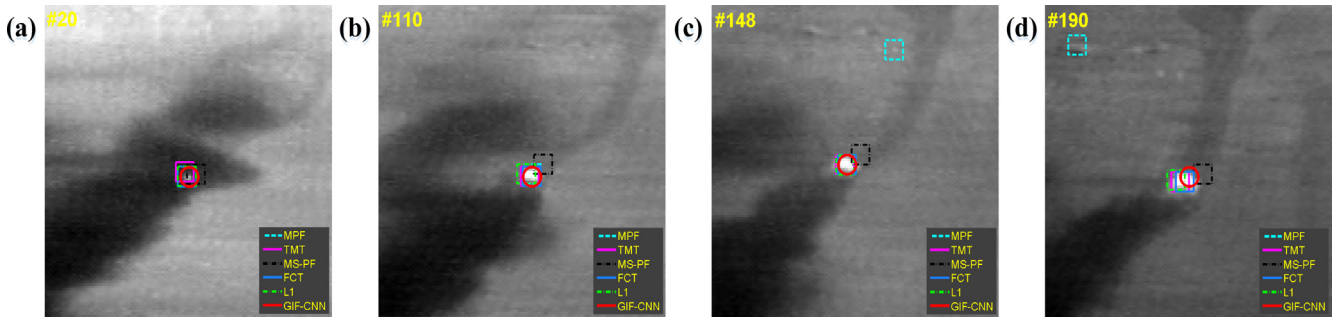


Fig. 15. Qualitative results of the 6 trackers over *tank* with ground background. Best viewed on color display.

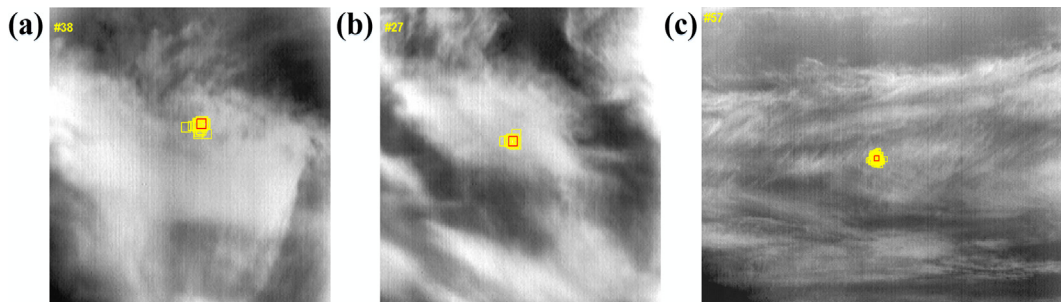


Fig. 16. Illustration of sampled particles. (a) *plane1*, (b) *plane2*, (c) *plane3*.

Particles' weights are obtained through the response of the online classifier, which trains some discriminative samples.

The GIF-CNN tracker has done well in tracking the dim moving target regardless of bright background or strong edge. The GIF algorithm alleviates the influence of edge-cloud in sampled templates, which is similar to the guidance, and blurs the negative templates when samples fails to match the target template. Besides, the particle filter is an efficient framework, reducing track-

ing deviation, and the online process of updating parameters in the native Bayesian classifier makes the algorithm adapt to the environment continuously.

4.3. Quantitative comparison

For further quantitative analysis, a similar standard evaluation in [19] is adopted, i.e., Center Location Error (CLE), Success Rate

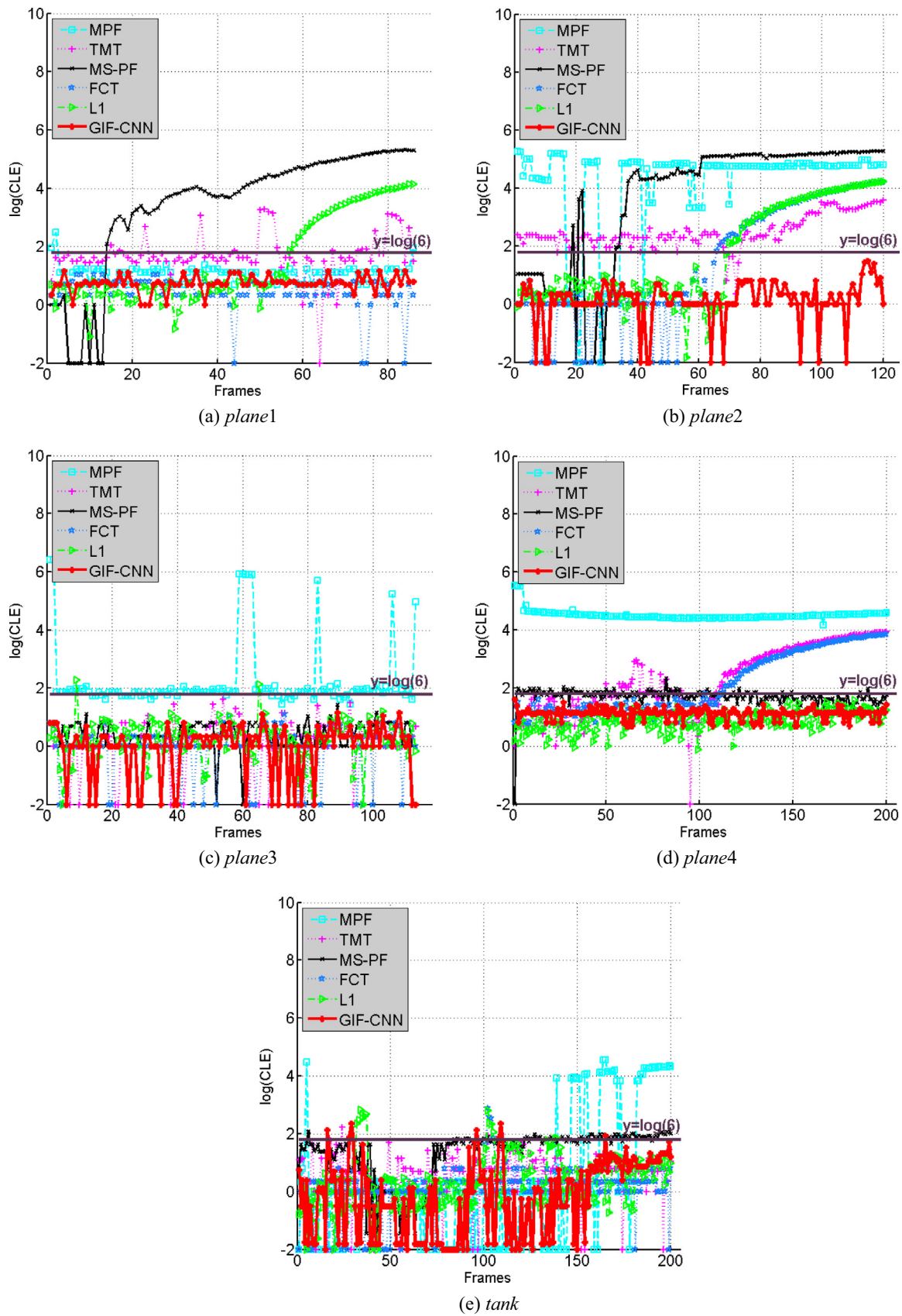


Fig. 17. Illustration of Center Location Error.

Table 2

Success Rate (SR) (%) and average frame per second (FPS). **Bold** fonts indicate the best performance while the *italic* fonts indicate the second best ones. The total number of evaluated frames is 719.

Sequence	MPF	TMT	FCT	MS-PF	ℓ_1	GIF-CNN
<i>Plane1</i>	0.96	0.77	1	0.14	0.65	1
<i>Plane2</i>	0.12	0.08	0.54	0.24	<i>0.58</i>	0.95
<i>Plane3</i>	0.65	0.97	1	1	<i>0.98</i>	1
<i>Plane4</i>	0	0.47	0.56	<i>0.91</i>	1	1
<i>Tank</i>	0.81	0.99	0.98	0.91	0.94	<i>0.98</i>
Average SR	0.51	0.66	0.82	0.64	<i>0.83</i>	0.98
Average FPS	5	24	40	20	12	19

(SR) and Frame per Second (FPS). The CLE denotes the Euclidean distance d between the estimated target and the real target. The SR is defined as n_1/m , where m is the number of all frames, and n_1 denotes the frame number on condition that $d < 6$. Thus, the CLE and the SR represent algorithm accuracy while the FPS represents real-time performance. Then, a group of comparison results is shown in Fig. 17 and Table 2.

In Fig. 17, the vertical axis represents the logarithm of the CLE value, for a suitable scale. $\log$ represents the logarithm operator and the line ' $y = \log(6)$ ' is a boundary (equal to that the CLE value is 6). As shown in Fig. 17(d), the CLE curve of the MPF algorithm over the *plane4* sequence is relatively unacceptable. In Fig. 17(a), the MS-PF algorithm undergoes a large fluctuation after frame #15, due to the fact that the target moves into the bright cloud. Besides, the ℓ_1 algorithm drifts slowly to the background after frame #60 and tracking deviation is unrecoverable. In Fig. 16(b), only the GIF-CNN algorithm obtains a satisfactory estimation at most frames and other algorithms lose the target in dozens of frames, for the reason that the *plane2* sequence contains strong edge and targets with low SNR. With severe background clutter, the target template in the TMT algorithm cannot be updated ideally, leading to the cumulative error, which is shown in Figs. 17(b) and (d). In addition, the false alarm rate of the MPF algorithm increases largely (see Fig. 17(b)), because it regards the moving edge as a target easily. It is indicated in Fig. 17(d) that the TMT algorithm always loses the target and the PF-based algorithms (the MS-PF and GIF-CNN algorithms) can deal with the cluttered background well. Moreover, both the TMT algorithm and the FCT algorithm are influenced by false targets and corresponding deviation is unrecoverable (see Fig. 17(d)). In the *tank* sequence, camera shake makes the target unstable. As shown in Fig. 17(e), the CLE results of several algorithms (except the MPF algorithm) are satisfactory with sufficient target information. Yet, the presented algorithm shows a good performance of tracking a small target over all infrared sequences.

As shown in Table 2, the success rate result of the MPF algorithm over the *plane4* sequence are 0, due to the fact that this algorithm is sensitive to the environment. For the TMT algorithm, tracking deviation in the matching process results in an unrecoverable detection (see Figs. 17(b) and (d)) and its success rate over the *plane2* sequence is 0.08. The MPF algorithm has a modest success rate (up to 0.96), yet, it is time-consuming. Specially, its FPS is 5 and the value can be smaller as the image size becomes larger. Besides, it is greatly influenced by the edge (see Fig. 17(b)) and the success rate over Seq.2 decreases to 0.12. Since the MS-PF algorithm utilizes the gradient histogram feature of an infrared image template, it cannot obtain a stable gray statistics while the target submerges into the bright cloud (e.g., Seq.1 and Seq.2). For example, in the *plane1* sequence, its success rate (0.14) is much smaller than that in Seq.3 ~ 6. The FCT algorithm has a low success rate if the target moves into in the cluttered background (e.g., Seq.2), resulting in an inaccurate classification.

The GIF-CNN algorithm does well in tracking a dim moving target over kinds of complex backgrounds. On the one hand, the GIF algorithm makes candidate templates be more discriminative. On the other hand, the CNN model obtains sufficient target information even if the target has a low gray value (e.g., *plane2* and *plane4*) or undergoes severe background (e.g., *plane1*). Experimental results insist that the presented algorithm is superior to the other five algorithms in the accuracy.

5. Conclusions

In this paper, an infrared dim and small target tracking algorithm based on the guide filter and the convolutional network is presented. Extracted templates can be filtered to be either smooth due to high similarity with the guidance image, or blurry because of the mismatch with the guidance. Moreover, the powerful two-layer convolutional network method is utilized to obtain effective template representation. The appearance model helps to generate a robust tracker based on the native Bayesian classification, which is adopted to obtain particles' weights. Besides, a 2-order transition model which considers two previous statuses is adopted to estimate the current position for each particle. Finally, experimental results demonstrate that the presented algorithm outperforms relative algorithms in the accuracy, and broadly, makes great contributions on laying the foundation for future recognition.

Conflict of interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service or company that could be construed as influencing the review of the manuscript entitled 'Guided filter and convolutional network based tracking for infrared dim moving target'.

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References

- [1] X. Wang, C. Ning, L.Z. Xu, Spatiotemporal difference-of-Gaussians filters for robust infrared small target tracking in various complex scenes, *Appl. Opt.* 54 (7) (2015) 1573–1586.
- [2] C.L.P. Chen, H. Li, Y.T. Wei, et al., A local contrast method for small infrared target detection, *IEEE Trans. Geosci. Remote Sens.* 52 (1) (2014) 574–581.

- [3] R.M. Liu, E.Q. Liu, J. Yang, et al., Automatically detect and track infrared small targets with kernel Fukunaga-Koontz transform and Kalman prediction, *Appl. Opt.* 46 (31) (2007) 7780–7791.
- [4] W. Wang, Z. Niu, Z. Chen, Temporal-spatial fusion filtering algorithm for small infrared moving target detection, *Infrared Laser Eng.* 34 (6) (2005) 714–718.
- [5] J. Liu, H. Ji, Detection algorithm for small targets in the IR image based on the variable weighted pipeline filter, *J. Xidian Univ.* 34 (2007) 743–747.
- [6] V.H. Diaz-Ramirez, K. Picos, V. Kober, Target tracking in nonuniform illumination conditions using locally adaptive correlation filters, *Opt. Commun.* 323 (2014) 32–43.
- [7] V.H. Diaz-Ramirez, V. Contreras, V. Kober, et al., Real-time tracking of multiple objects using adaptive correlation filters with complex constraints, *Opt. Commun.* 309 (2013) 265–278.
- [8] L. Chang, Z.H. Liu, S.T. Wu, Tracking of infrared radiation dim target based on mean-shift and particle filter, in: *IEEE transaction Guidance, Navigation and Control Conference*, 2014, pp. 671–675.
- [9] J.F. Dou, J.X. Li, Robust visual tracking based on joint multi-feature histogram by integrating particle filter and mean shift, *Optik-Int. J. Light Electron Opt.* 126 (15) (2015) 1449–1456.
- [10] F. Wang, E. Liu, J. Yang, S. Yu, et al., Target tracking in infrared imagery using a novel particle filter, *Chin. Opt. Lett.* 7 (7) (2009) 576–579.
- [11] Z.Z. Li, J.N. Li, F.Z. Ge, et al., Dim moving target tracking algorithm based on particle discriminative sparse representation, *Infrared Phys. Technol.* 75 (2016) 100–106.
- [12] R.M. Liu, X.L. Li, L. Han, et al., Track infrared point targets based on projection coefficient templates and non-linear correlation combined with Kalman prediction, *Infrared Phys. Technol.* 57 (2013) 68–75.
- [13] R.M. Liu, Y.H. Lu, C.L. Gong, et al., Infrared point target detection with improved template matching, *Infrared Phys. Technol.* 55 (4) (2012) 380–387.
- [14] S. Baker, I. Matthews, Lucas-Kanade 20 years on: a unifying framework, *Int. J. Comput. Vision* 56 (3) (2004) 221–255.
- [15] L. Liu, Z. Huang, Improved motion information-based infrared dim target tracking algorithms, *Infrared Phys. Technol.* 67 (2014) 341–349.
- [16] Z. Wang, Q. Hou, L. Hao, Improved infrared target-tracking algorithm based on mean shift, *Appl. Opt.* 51 (21) (2012) 5051–5059.
- [17] I. Leichter, Mean shift trackers with cross-bin metrics, *IEEE Trans. Pattern Anal. Mach. Intelligence* 34 (4) (2012) 695–706.
- [18] J. Ling, E. Liu, H. Liang, et al., Infrared target tracking with kernel-based performance metric and eigenvalue-based similarity measure, *Appl. Opt.* 46 (16) (2007) 3239–3252.
- [19] K.H. Zhang, L. Zhang, M.H. Yang, Fast compressive tracking, *IEEE Trans. Pattern Anal. Mach. Intell.* 36 (10) (2014) 2002–2015.
- [20] Y. Li, P.C. Li, Q. Shen, Real-time infrared target tracking based on ℓ_1 minimization and compressive features, *Appl. Opt.* 53 (28) (2014) 6518–6526.
- [21] L. Wang, T. Liu, G. Wang, et al., Video tracking using learned hierarchical features, *IEEE Trans. Image Process.* 24 (4) (2015) 1424–1435.
- [22] K.H. Zhang, Q.S. Liu, Y. Wu, et al., Robust visual tracking via convolutional networks without training, *IEEE Trans. Image Process.* 25 (4) (2016) 1779–1792.
- [23] K.M. He, J. Sun, X.O. Tang, Guided image filtering, *IEEE Trans. Pattern Anal. Mach. Intelligence* 35(6) (2013) 1397–1409.
- [24] M.S. Arulampalam, S. Maskell, N. Gordon, et al., A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking, *IEEE Trans. Signal Process.* 50 (2) (2002) 174–188.
- [25] B. Babenko, M.H. Yang, S. Belongie, Robust object tracking with online multiple instance learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 33 (8) (2011) 1619–1632.
- [26] D.D. Du, H.C. Lu H, L.H. Zhang, et al., Visual tracking via guided filter, in: *IEEE International Conference on Image Processing*, 2015, pp. 1781–1785.
- [27] J.W. Davis, OTCBVS benchmark dataset collection, (2007) <<http://www.cse.ohio-state.edu/otcbvs-bench/>>.